HumanOS

PROJECT PROPOSAL

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1. PROJECT OVERVIEW

1.1 WHAT IT IS

Just like a computer manages the hardware, HumanOS manages your focus. It is like a productivity assistant that works in the background to help you focus to use the computer to the maximum efficiency.

1.2 HUMANOS BASICS

The HumanOS works to track your input, use a smart algorithm to decide how focused you are and then give suggestions on how to improve your focus. The model learns how you focus and adjusts its weights to match with your focus intervals and your focus patterns, such as typing, to understand how you work to maximize your efficiency. It monitors your focus and allows you to see how you focus over time in a simplistic UI.

1.3 EXAMPLE

For example, a user is getting distracted while watching a YouTube video and slows their typing, sometimes idling for a minute. The HumanOS realizes that the user is distracted based on their patterns and suggests that the user close the window. The user clicks yes to the suggestion and the window is closed.

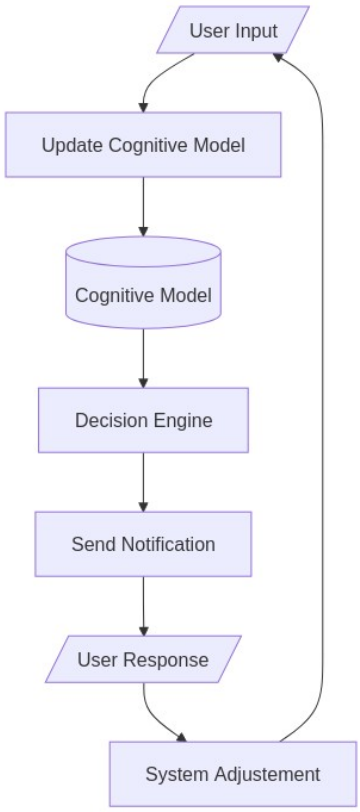
2. WHY CHOSE THIS PROJECT

I chose this project because I have always been an advocate for human-computer synergy and using computers in a productive way instead of wasting time scrolling the internet. I asked ChatGPT how I could enhance this almost human enhancement idea of mine and it suggested that I made a background functional productivity app that helps use the computer to maximum efficiency and guide focus. For example, a user could be slowing down a lot on their typing and eventually begin idling for long periods of time. The system would suggest a short break for the user to recharge before returning to the computer. I thought this was a great idea and using a system tray and making it fully functional by itself while asking user suggestions before committing changes to the users flow was an amazing idea and one that I have never seen before. I believe that this will help at least me focus better on my schoolwork and be more productive and I enjoy this as a concept.

3. FEATURES

SYSTEM TRACKING	The system tracks keyboard, mouse, and window switches to detect patterns to deduce focus.
SYSTEM TRAY ICON	A icon will be in the system tray that changes colors and contains information related to your focus.
SMART SUGGESTIONS	The tray icon will pop up a notification when it has a suggestion. It will change its icon to have a bell next to it with a tiny satisfying notification sound to notify you of a suggestion. The goal is for it to not be annoying and invasive, only helpful.
FOCUS BATTERY	A focus battery score will be displayed in the tray icon so that you can know how focused you are when you actually are at any given moment.
ADAPTIVE ALGORITHMS	The suggestions are adaptive to each situation depending on the user input and user focus score. Combining the two gives suggestions determined by the situation to improve focus.
PREEMPTIVE SUGGESTIONS	Suggestions are given before energy dips to make sure that time spent working is fully efficient.
NOTIFICATIONS	Notifications are given in the system tray icon to show suggestions. The notifications are meant to be calm, non-invasive, and completely optional.
USER PERSONALIZATION	The system logs the user input and focus and learns their patterns over time. It stores these biases as weights to make sure algorithms determine how user is most efficient. Maybe they spend less time typing and more time thinking so keyboard input would have less weight.
GRAPH OF FOCUS	A small graph will be displayed in the system tray icon to show focus over time for the user to see.
IDLE DETECTION	The system has smart idle detection to tell when the user steps away from the desktop. It pauses the focus system to not skew the algorithm and decrement focus score in the case of breaks/bathroom breaks.

4. ARCHITECTURE

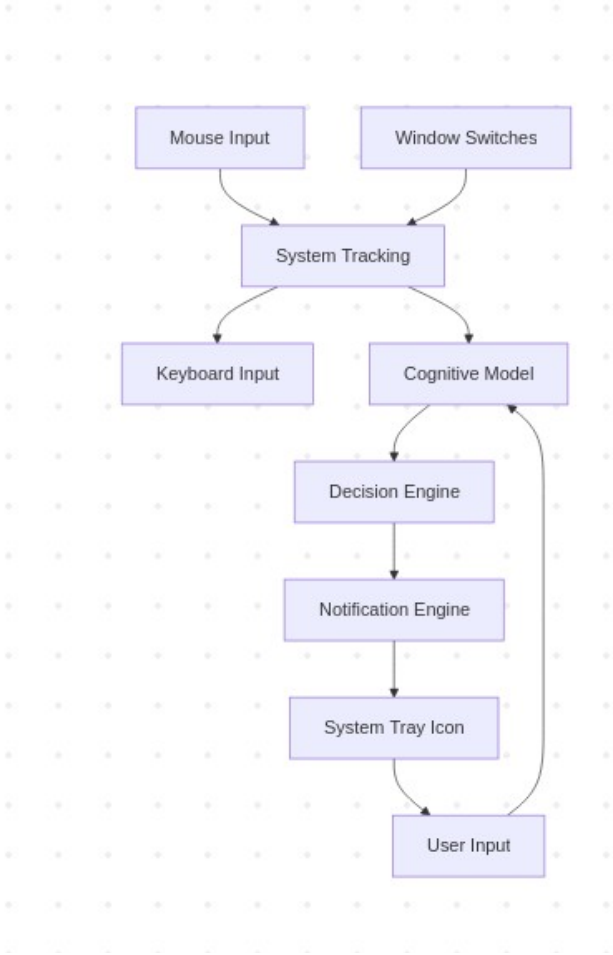


The user input is tracked and used to update the cognitive model. The decision engine then uses the user input and the cognitive model to judge if there is any action the user could take to improve focus. If there is an action the user could take, the system then notifies the user a suggestion. The user can then choose to accept the action or not and the system will adjust unless it is an action only the user can take, for example, taking a short break. The cycle then repeats, with suggestions only being given every once in awhile to be helpful, not invasive.

5. SOFTWARE MODULES

TRACKING SYSTEM	The system tracks keyboard, mouse, and window switches to detect patterns to deduce focus.
SYSTEM TRAY ICON	A icon will be in the system tray that changes colors and contains information related to your focus.
COGNITIVE MODEL	The tray icon will pop up a notification when it has a suggestion. It will change its icon to have a bell next to it with a tiny satisfying notification sound to notify you of a suggestion. The goal is for it to not be annoying and invasive, only helpful.
DECISION MODEL	A focus battery score will be displayed in the tray icon so that you can know how focused you are when you actually are at any given moment.
NOTIFICATION MODULE	The suggestions are adaptive to each situation depending on the user input and user focus score. Combining the two gives suggestions determined by the situation to improve focus.
USER INTERFACE	Suggestions are given before energy dips to make sure that time spent working is fully efficient.

6. DATAFLOW



The system tracker takes input and pipes it to the cognitive model that categorizes how focused the user is. The system data and cognitive model are then used in the decision-making process and rules to send notifications. The notification suggestion pop up in the system tray is either accepted or denied by the user and the system automatically adjusts if needed. The GUI system icon color is updated. The cycle continues as needed to continue to assist the user in maximum focus

7. TIMELINE AND MILESTONES

SPRING BREAK	Finish tray icon.
END OF MARCH	Finish decision tree and notifications plus weights.
QUARTER APRIL	Finish UI in icon tray to see activity over time graph and notification settings.
END OF APRIL	Finish system actions (closing tabs by itself). Add weights to algorithms.
MAY	Demo and presentation.

8. CONCLUSION

I am excited to pursue this project and show my passion for non-addictive focus software that will help instead of harm. I have set realistic goals for myself that are achievable by the end of the semester. My effort will prepare a brilliant program with an amazing demonstration by the time May comes. I encourage you to checkout my progress on the project on GitHub at github.com/zach-richards/human-os.